

# Motility-induced phase separation

When slowing down causes clustering

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Self-propelled particles can accumulate without an attractive potential. The simplest feedback is kinetic: particles remain longer where they move slowly, and local crowding can reduce their speed further. A one-dimensional run-and-tumble model makes the instability particularly transparent.

## 1 Eliminating the directional imbalance

Let  $R(x, t)$  and  $L(x, t)$  be densities moving right and left at positive speed  $v$ . Each particle switches direction at rate  $\alpha/2$ , so the directional imbalance relaxes at rate  $\alpha$ . Neglect translational Brownian diffusion, alignment, birth, death, and fluid-mediated interactions:

$$\partial_t R = -\partial_x(vR) - \frac{\alpha}{2}R + \frac{\alpha}{2}L, \quad (1)$$

$$\partial_t L = +\partial_x(vL) + \frac{\alpha}{2}R - \frac{\alpha}{2}L. \quad (2)$$

Writing  $\rho = R + L$  and  $p = R - L$  gives

$$\partial_t \rho = -\partial_x(vp), \quad \partial_t p = -\partial_x(vp) - \alpha p. \quad (3)$$

On times long compared with  $\alpha^{-1}$  and lengths large compared with the persistence length  $v/\alpha$ , adiabatic elimination gives

$$p \simeq -\frac{1}{\alpha}\partial_x(vp), \quad J = vp \simeq -\frac{v}{\alpha}\partial_x(vp). \quad (4)$$

For prescribed spatially varying  $v(x)$ , a zero-current steady state satisfies  $vp = \text{constant}$ , so  $\rho \propto 1/v$ . Motility affects a steady distribution directly, unlike a passive particle's mobility in thermal equilibrium with a fixed potential.

## 2 Density-dependent speed creates feedback

Now adopt a local mean-field constitutive law  $v = v(\rho)$ . Equation (4) becomes

$$J = -D_{\text{coll}}(\rho)\partial_x\rho, \quad D_{\text{coll}} = \frac{v^2}{\alpha} \left(1 + \rho \frac{v'}{v}\right). \quad (5)$$

For  $\rho = \rho_0 + \delta\rho e^{st+iqx}$ , linearization yields  $s = -D_{\text{coll}}(\rho_0)q^2$ . The homogeneous state is unstable to long wavelengths when

$$\boxed{\frac{d \ln v}{d \ln \rho} < -1.} \quad (6)$$

An excess of particles then reduces the outgoing flux enough to amplify itself. This is a spinodal criterion in the stated closure, not a complete phase diagram. Adding a constant translational diffusivity  $D_t$  adds  $D_t$  to  $D_{\text{coll}}$  and can suppress the instability.

### 3 A limited bridge to equilibrium phase separation

Introduce fixed positive reference scales  $\rho_*$  and  $v_*$ . The same deterministic flux can be written

$$J = -M(\rho)\partial_x\mu, \quad M(\rho) = \frac{\rho v(\rho)^2}{\alpha}, \quad \mu = \ln \frac{\rho}{\rho_*} + \ln \frac{v(\rho)}{v_*}. \quad (7)$$

For a speed that depends only on the local density, define

$$\mathcal{F}[\rho] = \int f(\rho) dx, \quad (8)$$

$$f(\rho) = \rho \left( \ln \frac{\rho}{\rho_*} - 1 \right) + \int_0^\rho \ln \frac{v(s)}{v_*} ds. \quad (9)$$

Then  $\mu = \delta\mathcal{F}/\delta\rho$  and

$$f''(\rho) = \frac{1}{\rho} + \frac{v'}{v}, \quad \frac{d\mathcal{F}}{dt} = - \int M(\partial_x\mu)^2 dx \leq 0 \quad (10)$$

for periodic or no-flux boundaries. The spinodal is exactly the region  $f'' < 0$ . This mathematical Lyapunov functional provides an equilibrium-like description of the local closure; it is not automatically the physical free energy or entropy production of a swimming organism.

### 4 Why the short-wavelength physics matters

Negative diffusion alone makes the truncated equation unstable at arbitrarily large  $q$ , where the elimination leading to Eq. (4) is invalid. A simple illustrative regularization is

$$\mathcal{F} = \int \left[ f(\rho) + \frac{\kappa}{2} (\partial_x \rho)^2 \right] dx, \quad \mu = f'(\rho) - \kappa \partial_x^2 \rho, \quad \kappa > 0. \quad (11)$$

The resulting conserved dynamics has dispersion

$$s(q) = -M(\rho_0)q^2 \left[ f''(\rho_0) + \kappa q^2 \right]. \quad (12)$$

When  $f'' < 0$ , the fastest-growing linear mode has  $q^2 = -f''/(2\kappa)$ . This is an initial length scale, not necessarily the final domain size: conserved phase separation can continue to coarsen.

As a local example,  $v = v_0(1 - \rho/\rho_{\max})$  gives an instability for  $\rho > \rho_{\max}/2$  within  $0 < \rho < \rho_{\max}$ . Near vanishing speed, additional packing and persistence effects demand a better model. Coexistence densities cannot be inferred from this threshold alone.

Run-and-tumble bacteria motivate the kinetic picture [1]. In biological populations, chemotaxis, growth, adhesion, and hydrodynamics can also cause clustering. Moreover, gradient corrections in active systems need not derive from a functional: an equilibrium common-tangent construction is justified only when the required mapping actually holds [2].

## References

- [1] J. Tailleur and M. E. Cates, “Statistical Mechanics of Interacting Run-and-Tumble Bacteria” (2008). [arXiv:0803.1069](#).
- [2] M. E. Cates and J. Tailleur, “Motility-Induced Phase Separation,” *Annu. Rev. Condens. Matter Phys.* **6**, 219–244 (2015). [arXiv:1406.3533](#).